

Notes: Blanchard, Bown, Johnson (RES 2026)

Global Value Chains and Trade Policy

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1 Executive overview

- Core question: how do **global value chain (GVC)** linkages change a government's incentives to impose **import protection** (tariffs and temporary trade barriers)?
- Key modelling idea: GVCs are a form of **trade in factor services**. They break the link between:
 1. *where* a final good is produced (the location where tariffs are applied), and
 2. *who* earns the income generated by that production (value added is earned by factor owners across countries).
- Main theoretical prediction: discretionary final-goods tariffs fall when:
 1. imports contain more **domestic value added** (my country is upstream in foreigners' exports), and
 2. my own final-goods production contains more **foreign value added** (protectionist rents leak to foreign upstream suppliers).
- Empirics: using 14 countries (incl. EU as a region) and WIOD value-added content measures (1995–2015), they show lower **tariffs** and lower **TTB** use where GVC linkages are stronger.
- Quantitative counterfactuals: severing GVCs makes **tariff preferences** (applied < MFN) largely disappear; decoupling China from G7 upstream inputs raises optimal G7 tariffs on Chinese final goods.

2 Notation

- Home vs Foreign (or many countries later). We denote Home by h , Foreign by f , and generic country by c .
- Final good: x (the traded non-numeraire); numeraire: y (freely traded).
- Final-goods tariff: $\tau \geq 1$ on imports of x (so tariff rate is $t \equiv \tau - 1$).
- Key GVC objects:
 - FVA_h : **foreign value added** embodied in Home production of x (foreign upstream inputs used in domestic production).
 - DVA_h : **domestic value added** embodied in Foreign production of x (Home upstream inputs used in foreign production).
- Empirics uses ratios of these objects to gross bilateral imports.

3 Theory: the 2×2 benchmark (specific factors as GVC inputs)

3.1 Environment

Countries and goods. Two countries $c \in \{h, f\}$ and two goods $s \in \{x, y\}$. Let y be the numeraire and p^c be the local relative price of x in country c .

Preferences. Aggregate utility in country c :

$$U(d_x^c, d_y^c),$$

with d_s^c total consumption of good s .

Factors.

- Homogeneous labor (mobile across sectors, immobile across countries).
- **GVC inputs** that are specific (in the benchmark) to sector and location of use.

Let ν_h^c and ν_f^c be quantities of Home- and Foreign-origin GVC inputs used in production of final good x in country c . Define $\vec{\nu}^c \equiv (\nu_h^c, \nu_f^c)$.

Technology.

$$q_x^c = f_x^c(l_x^c, \nu_h^c, \nu_f^c), \quad q_y^c = l_y^c. \quad (1.1)$$

Perfect competition. With y produced by labor alone, the wage is normalized to 1.

Tariff. Home imposes an ad-valorem tariff τ on imports of final good x (applied to the foreign selling price).

3.2 Equilibrium objects (production, bargaining, income)

Labor allocation. Given p^c and $\vec{\nu}^c$:

$$l_x^c(p^c; \vec{\nu}^c) \in \arg \max_{l_x^c} \{p^c f_x^c(l_x^c, \nu_h^c, \nu_f^c) - l_x^c\}. \quad (1.2)$$

$$l_y^c(p^c; \vec{\nu}^c) = L^c - l_x^c(p^c; \vec{\nu}^c). \quad (1.3)$$

Supplies:

$$q_x^c(p^c; \vec{\nu}^c) = f_x^c(l_x^c(p^c; \vec{\nu}^c); \vec{\nu}^c), \quad q_y^c(p^c; \vec{\nu}^c) = l_y^c(p^c; \vec{\nu}^c). \quad (1.4-1.5)$$

Residual profit and GVC rents. Residual profit from producing x in country c :

$$\pi_x^c(p^c; \vec{\nu}^c) \equiv p^c q_x^c(p^c; \vec{\nu}^c) - l_x^c(p^c; \vec{\nu}^c).$$

The paper assumes residual profits are split among GVC input suppliers via a weighted Nash bargaining solution:

$$\pi_{xd}^c(p^c; \vec{\nu}^c) = \alpha_d^c \pi_x^c(p^c; \vec{\nu}^c), \quad \sum_d \alpha_d^c = 1, \quad \alpha_d^c > 0.$$

Return per unit to origin- d GVC input used in country c :

$$r_d^c(p^c; \vec{\nu}^c) \equiv \frac{\pi_{xd}^c(p^c; \vec{\nu}^c)}{\nu_d^c}.$$

Key qualitative property: r_d^c increases in p^c (downstream prices pass through to upstream incomes).

Demand and indirect utility.

$$d_x^c(p^c, I^c) \in \arg \max_{d_x^c} \{U(d_x^c, d_y^c) : d_y^c + p^c d_x^c \leq I^c\}, \quad (1.6)$$

$$d_y^c(p^c, I^c) = I^c - p^c d_x^c(p^c, I^c), \quad (1.7)$$

$$V(p^c, I^c) = U(d_x^c(p^c, I^c), d_y^c(p^c, I^c)). \quad (1.8)$$

Income decomposition and the GVC terms. The paper emphasizes an accounting that isolates foreign and domestic value-added components:

$$I^c = p^c q_x^c(p^c; \vec{\nu}^c) + q_y^c(p^c; \vec{\nu}^c) + R^c - FVA^c + DVA^c. \quad (1.10)$$

Where (with $j \neq c$):

$$FVA^c \equiv r_j^c(p^c; \vec{\nu}^c) \nu_j^c, \quad DVA^c \equiv r_c^j(p^j; \vec{\nu}^j) \nu_c^j. \quad (1.11)$$

Interpretation: FVA^c is income generated domestically that accrues to foreign upstream suppliers; DVA^c is income earned by domestic upstream suppliers from foreign production.

3.3 Prices, trade, and the tariff link

No-arbitrage implies:

$$p^h(\tau, p^f) = \tau p^f. \quad (1.12)$$

Foreign price is pinned down by market clearing in x :

$$d_x^h(p^h(\tau, p^f), p^f; \vec{\nu}) + d_x^f(p^h(\tau, p^f), p^f; \vec{\nu}) = q_x^h(p^h(\tau, p^f); \vec{\nu}^h) + q_x^f(p^f; \vec{\nu}^f). \quad (1.13)$$

3.4 Government problem and the key optimal tariff condition

Home chooses τ to maximize indirect utility, internalizing equilibrium responses:

$$\tau^o \in \arg \max_{\tau} V(p^h, I(p^h, p^f)) \quad \text{s.t. } p^h = \tau p^f, \quad p^f = \tilde{p}^f(\tau). \quad (1.14)$$

After applying Roy's identity and rewriting, the first-order condition can be expressed as:

$$V_{\tau} = V_I \left[(\tau^o - 1) p^f \frac{dM_x}{d\tau} - M_x \frac{d\tilde{p}^f}{d\tau} - \frac{dFVA^h}{d\tau} + \frac{dDVA^h}{d\tau} \right] = 0. \quad (1.16)$$

Interpretation of (1.16) (this is the core intuition).

- The **first two terms** are the standard **terms-of-trade** (cost shifting) motive: a large importer wants to push down the foreign price.
- The last two are **GVC terms**:
 - $dDVA^h/d\tau < 0$: by pushing down the foreign output price, Home hurts *itself* because its own upstream factors are embedded in foreign exports (domestic content in imports). That dampens the incentive to raise tariffs.
 - $dFVA^h/d\tau > 0$: by raising the local domestic price, Home creates protection rents, but if domestic production uses foreign upstream inputs, those rents leak abroad; that also dampens tariffs.

Pass-through elasticities into GVC income. The paper rewrites the derivatives as elasticities:

$$\frac{dFVA^h}{d\tau} = \varepsilon_{r_f^h} \frac{FVA^h}{p^h} \frac{d\tilde{p}^h}{d\tau} > 0, \quad \frac{dDVA^h}{d\tau} = \varepsilon_{r_h^f} \frac{DVA^h}{p^f} \frac{d\tilde{p}^f}{d\tau} < 0, \quad (1.17-1.18)$$

where $\varepsilon_{r_f^h}$ and $\varepsilon_{r_h^f}$ are elasticities of upstream returns with respect to local downstream prices.

3.5 Optimal tariff formula with GVC linkages

The benchmark expression can be summarized as:

$$\tau^o = 1 + \frac{1}{\varepsilon_x^f} \left(1 - \varepsilon_{r_h^f} \frac{DVA^h}{p^f E_x^f} - \varepsilon_{r_f^h} \frac{FVA^h}{p^h E_x^f} \cdot \frac{1}{|\lambda|} \right), \quad \lambda \equiv \frac{d\tilde{p}^f/d\tau}{d\tilde{p}^h/d\tau} < 0. \quad (1.19)$$

Takeaway from (1.19).

- Without GVCs ($DVA = FVA = 0$), I get the classic inverse export supply elasticity rule:

$$\tau^o = 1 + \frac{1}{\varepsilon_x^f}.$$

- With GVCs, **both** more domestic content in imports (DVA^h) and more foreign content in domestic production (FVA^h) push the optimal tariff **down**.
- The magnitudes scale with trade volume and elasticities (because the terms are normalized by pE objects).

4 Extension: endogenous GVC reorganization

Endogenous $\vec{\nu}$ and the modified optimal tariff

They allow input allocations to respond to prices and adopt quasi-linear preferences. The optimal tariff becomes:

$$\tau^o = 1 + \frac{1}{\varepsilon_x^f} \left(1 - (\tilde{\varepsilon}_{r_h^f} + \tilde{\varepsilon}_{\nu_h^f}) \frac{DVA^h}{p^f E_x^f} - (\tilde{\varepsilon}_{r_f^h} + \tilde{\varepsilon}_{\nu_f^h}) \frac{FVA^h}{p^h E_x^f} + \eta \right). \quad (1.20)$$

Interpretation. Allowing endogenous reorganization adds:

- extra pass-through terms that reflect both **price** and **quantity** responses of inputs,
- an additional term η capturing general equilibrium reallocation effects.

The key qualitative prediction (tariffs decreasing in DVA and FVA) continues to hold under conditions ensuring that Home's GVC income falls when it raises tariffs (i.e., the value chain is sufficiently segmented so local prices matter most for local input earnings).

5 Input tariffs (why they mostly set them aside)

Exogenous input tax g on foreign-sourced GVC inputs used in Home

They introduce an exogenous ad-valorem input tax $g \in [0, 1]$ on foreign inputs used at Home (conceptually: a tax on foreign value added in domestic production). Tariff revenue becomes:

$$R^h = (p^h - p^f)M_x + g r_f^h \nu_f^h. \quad (1.21)$$

FOC becomes:

$$V_\tau = V_I \left[(\tau^o - 1) p^f \frac{dM_x}{d\tau} - M_x \frac{d\tilde{p}^f}{d\tau} - (1 - g) \frac{dFVA^h}{d\tau} + \frac{dDVA^h}{d\tau} \right] = 0. \quad (1.22)$$

Optimal final-goods tariff:

$$\tau^o = 1 + \frac{1}{\varepsilon_x^f} \left(1 - \varepsilon_{r^f} \frac{DVA^h}{p^f E_x^f} - (1 - g) \varepsilon_{r^f} \frac{FVA^h}{p^h E_x^f} \cdot \frac{1}{|\lambda|} \right). \quad (1.23)$$

Interpretation. An input tax partially “captures” the rents that would otherwise leak to foreign upstream suppliers, so it attenuates the FVA channel in the optimal *final-goods* tariff. But the directional predictions for DVA and FVA effects on final-goods tariffs remain.

Endogenous input tariffs

In the specific-factors benchmark, optimal input tariffs become corner ($g^o = 1$), which is unrealistic empirically (input tariffs are typically low). In more general settings, the optimal input tariff condition includes many cross-elasticities and becomes model-dependent (so they focus empirically on final-goods protection).

6 Many-country political-economy model + mapping to data

6.1 Preferences (many goods)

For Home h with many final goods $s \in S$ plus numeraire good 0:

$$U^h(d_0^h, \vec{d}^h) = d_0^h + \sum_{s \in S} u_s(d_s^h). \quad (2.1)$$

6.2 Government objective with special interests

They add political weights on rents (“protection for sale” logic):

$$G^h = V^h + \sum_{s \in S} \left[\delta_s^{DPE} \pi_s^h + \delta_s^{DVA} DVA_s^h + \delta_s^{FVA} FVA_s^h \right]. \quad (2.2)$$

Interpretation of the δ ’s.

- δ_s^{DPE} : extra weight on domestic downstream producer rents (import-competing sector politics).
- δ_s^{DVA} : extra weight on domestic upstream suppliers that earn income abroad (domestic content embedded in foreign goods).
- δ_s^{FVA} : extra weight (if any) on foreign upstream suppliers earning income from Home production. If this is small (plausible), the FVA channel should reduce tariffs.

6.3 Politically optimal bilateral tariffs

The model yields an implicit expression for bilaterally optimal final-goods tariffs:

$$\tau_{xj}^h = 1 + \frac{1}{\varepsilon_{xh}^j} \left(1 + \frac{\delta_x^{DPE}}{|\lambda_{xj}^h|} \cdot \frac{p_x^h q_x^h}{p_x^h E_{xh}^j} - (1 + \delta_x^{DVA}) \varepsilon_{xh}^{rj} \frac{DVA_{xh}^j}{p_x^j E_{xh}^j} - (1 - \delta_x^{FVA}) \varepsilon_{x*}^{rh} \frac{FVA_x^h}{p_x^h E_{xh}^j} \cdot \frac{1}{|\lambda_{xj}^h|} - \tilde{\Omega}_{xj} \right). \quad (2.3)$$

How I parse (2.3).

- Outside the parentheses: $\frac{1}{\bar{\epsilon}_{xh}^j}$ is the classic market-power / inverse supply elasticity term.
- Inside:
 - The **import-penetration** / **political economy** term scales with $p_x^h q_x^h / (p_x^h E_{xh}^j)$: protection is “cheaper” in welfare terms when import penetration is low.
 - The **DVA term** is negative: if Home value added is embedded in exporter j ’s goods, tariffs on j hurt domestic upstream incomes, so the tariff falls.
 - The **FVA term** is negative if $\delta^{FVA} < 1$: protection rents leak abroad via foreign inputs in domestic production, so tariffs fall.
 - $\tilde{\Omega}_{xj}$ captures third-country diversion effects.

6.4 Trade institutions: MFN constraint and RTAs

MFN. Applied bilateral tariffs are constrained above by MFN tariffs:

$$\tau_{xj}^{h,applied} \leq \tau_x^{h,MFN}.$$

Thus **tariff preferences** are deviations below MFN, and are *censored above by 0*:

$$t_{xjt}^{h,applied} - t_{xt}^{h,MFN} \leq 0.$$

RTAs. RTAs can involve negotiated cooperation that may alter the mapping; empirically, they control for RTA presence and also explore regime heterogeneity (mostly in the supplement).

6.5 Empirical linearization and estimating equation

They linearize around a no-GVC benchmark and motivate regressions in terms of ratios:

$$t_{xj}^h = \frac{1}{\bar{\epsilon}_{xh}^j} + \gamma^{IP} \left(\frac{FG_x^h}{\bar{p}_x^j \bar{E}_{xh}^j} \right) + \gamma^{DVA} \left(\frac{DVA_{xh}^j}{\bar{p}_x^j \bar{E}_{xh}^j} \right) + \gamma^{FVA} \left(\frac{FVA_x^h}{\bar{p}_x^h \bar{E}_{xh}^j} \right) + \omega_{xhj}. \quad (2.4)$$

Operationally they use log ratios with realized imports in denominators and estimate:

$$t_{xjt}^{h,applied} - t_{xt}^{h,MFN} = \min\{t_{xjt}^h - t_{xt}^{h,MFN}, 0\}, \quad (2.5)$$

$$t_{xjt}^h - t_{xt}^{h,MFN} = \Phi_{xhjt} + \gamma^{FG} \ln \left(\frac{FG_{xt}^h}{IM_{xjt}^h} \right) + \gamma^{DVA} \ln \left(\frac{DVA_{xht}^j}{IM_{xjt}^h} \right) + \gamma^{FVA} \ln \left(\frac{FVA_{xt}^h}{IM_{xjt}^h} \right) + e_{xhjt}. \quad (2.6)$$

Sign predictions (taking the dependent variable as “preference”). Because $t^{applied} - t^{MFN} \leq 0$:

- $\gamma^{DVA} < 0$: higher domestic content in partner exports $\Rightarrow$ more negative preference $\Rightarrow$ lower applied tariff.
- $\gamma^{FVA} < 0$ if $\delta^{FVA} < 1$: higher foreign content in domestic output $\Rightarrow$ lower tariff.
- $\gamma^{FG} > 0$: larger domestic production relative to imports (lower import penetration) $\Rightarrow$ more protection $\Rightarrow$ preference moves toward 0.

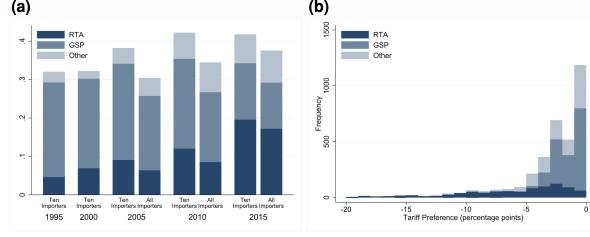


FIGURE 1
Tariff preferences. a) Tariff preferences over time and b) distribution of tariff preferences

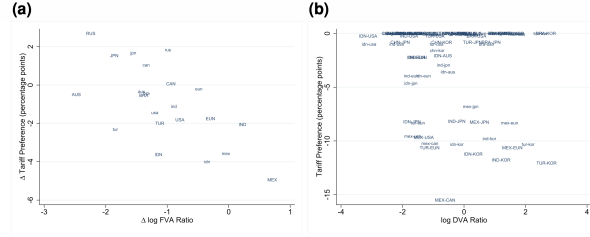


FIGURE 2
Textiles, leather, and footwear case study. a) Changes tariff preference versus log FVA ratio by importer and industry: 1995–2015 and b) tariff preferences versus log DVA ratio in 2015: high-income importers & emerging market exporters

7 Empirical facts: interpreting every figure and table in the paper

7.1 Figure 1: Tariff preferences in the data

What the figure shows.

- Figure 1(a): share of importer–exporter–industry cells (with nonzero MFN tariffs) that have preferential tariffs; broken down by regime (RTA, GSP, Other); plotted over 1995, 2000, 2005, 2010, 2015.
- Figure 1(b): histogram of *negative* tariff preferences (applied < MFN), excluding extreme preferences less than -20 percentage points for readability.

Interpretation.

- Preferences are **common** (roughly 30–40% of eligible cells), and they become **more prevalent** over time.
- “Depth” differs by regime: RTAs and similar agreements generate **larger-magnitude** preferences than GSP, so the shift toward RTAs deepens preferential liberalization.
- The distribution mass near 0 highlights the institutional censoring: many bilateral applied tariffs are at MFN (preference = 0).

7.2 Figure 2: Textiles/leather/footwear case study (two scatter plots)

What the figure shows.

- Figure 2(a): within an importer–industry cell, change in mean preferential tariff from 1995 to 2015 vs change in multilateral log FVA ratio $\Delta \ln(FVA/IM)$.
- Figure 2(b): cross section in 2015 for high-income importers vs emerging exporters: tariff preference vs $\ln(DVA/IM)$.

TABLE 1
Bilateral tariffs and value-added content

	OLS		Tobit	
	(1)	(2)	(3)	(4)
Panel A: Baseline				
DVA ratio: $\ln(DVA_{ijt}^j / IM_{ijt}^i)$	-1.00*** (0.27)	-1.08*** (0.30)	-2.39*** (0.51)	-2.53*** (0.53)
FVA ratio: $\ln(FVA_{ijt}^j / IM_{ijt}^i)$	-0.98*** (0.23)		-1.60*** (0.43)	
IP ratio: $\ln(FG_{ijt}^j / IM_{ijt}^i)$	2.16*** (0.49)		4.56*** (0.91)	
IP ratio + FVA ratio ($\gamma^{IP} + \gamma^{FVA}$)		1.27*** (0.32)		3.14*** (0.59)
Observations	11,385	11,385	11,385	11,385
R-Squared	0.364	0.387		
Panel B: Controlling for RTAs				
DVA ratio: $\ln(DVA_{ijt}^j / IM_{ijt}^i)$	-0.43** (0.17)	-0.47** (0.20)	-1.03*** (0.37)	-1.09*** (0.40)
FVA ratio: $\ln(FVA_{ijt}^j / IM_{ijt}^i)$	-0.71*** (0.19)		-1.16*** (0.34)	
IP ratio: $\ln(FG_{ijt}^j / IM_{ijt}^i)$	1.23*** (0.34)		2.44*** (0.66)	
IP ratio + FVA ratio ($\gamma^{IP} + \gamma^{FVA}$)		0.57*** (0.21)		1.36*** (0.44)
Reciprocal Trade Agreement: RTA_{ijt}	-4.35*** (0.60)	-4.30*** (0.61)	-7.72*** (1.06)	-7.63*** (1.05)
Observations	11,385	11,385	11,385	11,385
R-Squared	0.499	0.517		
Column Fixed Effects				
Importer-Year	Y	N	Y	N
Importer-Industry	Y	N	Y	N
Importer-Industry-Year	N	Y	N	Y
Exporter-Industry-Year	Y	Y	Y	Y

Interpretation.

- Panel (a) provides a **time-series** intuition: where foreign value added in domestic production becomes more important relative to imports, preferential tariffs expand more (tariffs are cut more).
- Panel (b) provides a **cross-sectional** intuition: when an importer has more of its own value added embedded in the exporter's goods (higher DVA ratio), it gives that exporter **lower applied tariffs**.
- The pile-up at 0 preference in (b) is the **MFN censoring** story that motivates Tobit in the regressions.

7.3 Table 1: Bilateral tariffs and value-added content (main reduced form)

Interpretation of Table 1.

- **DVA ratio effect:** negative and significant across OLS and Tobit. When the importer has *more of its own value added embedded in the exporter's final goods*, it offers **lower applied tariffs** (bigger preferences).
- **FVA ratio effect:** also negative and significant. When domestic production contains *more foreign value added*, the importer sets **lower tariffs** (consistent with protection rents leaking upstream abroad and/or weak political weight on foreigners).
- **Political economy control:** IP ratio is positive. A higher FG/IM (i.e., lower import penetration) is associated with **less negative** preferences (more protection), consistent with domestic producer influence.
- **MFN censoring:** Tobit coefficients are roughly **about twice** OLS in magnitude, which I read as evidence that MFN caps limit observed discriminatory tariff setting.
- **RTA controls (Panel B):** controlling for RTAs soaks up a lot of preference variation (RTA coefficient is large and negative), but DVA and FVA channels remain negative and significant.

TABLE 2
Heterogeneity by upstream and downstream differentiation

Panel A: Upstream differentiation				
	Manuf. versus Non-Manuf.		Rauch classification	
	OLS (1)	Tobit (2)	OLS (3)	Tobit (4)
DVA ratio (differentiated)	-1.20** (0.57)	-2.34** (0.97)	-2.40* (1.33)	-4.78** (2.31)
DVA ratio (non-differentiated)	0.11 (0.43)	-0.19 (0.92)	1.38 (1.20)	2.45 (2.28)
IP ratio + FVA ratio ($\gamma^{IP} + \gamma^{FVA}$)	1.26*** (0.31)	3.06*** (0.57)	1.20*** (0.28)	2.88*** (0.52)
Observations	11,385	11,385	11,385	11,385
R-Squared	0.395		0.398	
Panel B: Downstream differentiation				
	Manuf. versus Non-Manuf.		Rauch classification	
	OLS (5)	Tobit (6)	OLS (7)	Tobit (8)
DVA ratio (differentiated)	-1.09*** (0.29)	-2.64*** (0.54)	-1.27*** (0.32)	-1.27*** (0.29)
DVA ratio (non-differentiated)	-1.02*** (0.38)	-2.06*** (0.58)	-0.74** (0.29)	-0.74** (0.26)
IP ratio + FVA ratio (differentiated)	1.28*** (0.31)	3.22*** (0.59)	1.50*** (0.34)	1.50*** (0.31)
IP ratio + FVA ratio (non-differentiated)	1.25*** (0.42)	2.77*** (0.64)	0.90*** (0.31)	0.90*** (0.28)
Observations	11,385	11,385	11,385	11,385
R-Squared	0.387		0.392	

TABLE 3
Temporary trade barriers and value-added content

	(1)	(2)	(3)	(4)
DVA ratio	-0.22*** (0.061)	-0.19*** (0.066)		
DVA ratio (exporter = China)			-0.72*** (0.25)	-0.65*** (0.24)
DVA ratio (exporter $\neq$ China)			-0.16*** (0.054)	-0.14** (0.061)
FVA ratio	-1.28*** (0.39)		-1.28*** (0.39)	
IP ratio	1.46*** (0.40)		1.43*** (0.40)	
IP ratio + FVA ratio		0.18*** (0.057)		0.14*** (0.052)
Observations	10,098	10,098	10,098	10,098
R-Squared	0.235	0.477	0.239	0.480
Column fixed effects				
Importer-Year	Y	N	Y	N
Importer-Industry	Y	N	Y	N
Importer-Industry-Year	N	Y	N	Y
Exporter-Industry-Year	Y	Y	Y	Y

7.4 Table 2: Heterogeneity by differentiation (pass-through and elasticities)

Interpretation of Table 2.

- **Panel A (upstream differentiation):** DVA matters *only* when it comes from **differentiated upstream** sectors. DVA originating in non-differentiated upstream activities is not robustly related to tariffs. I read this as evidence that upstream differentiation proxies for **stronger pass-through** from downstream final-good prices to upstream factor returns (higher effective ε_r).
- **Panel B (downstream differentiation):** DVA is negative in both differentiated and non-differentiated downstream sectors, but tends to be **more negative** in differentiated downstream sectors (consistent with lower export supply elasticities in differentiated goods, so the DVA channel should have more bite).

7.5 Table 3: Temporary Trade Barriers (TTBs) and value-added content

Interpretation of Table 3.

- Dependent variable is a **coverage ratio** (share of imports covered by TTBs), so a negative coefficient means **less** TTB protection when GVC linkages are stronger.
- **DVA is negative:** if my country supplies more value added to the exporter's final goods, I am less likely to restrict those imports via TTBs.

- **China interaction:** the DVA channel is **much stronger** when China is the exporter. I interpret this as: countries avoid TTBS on China precisely when they have significant upstream domestic exposure in Chinese exports (i.e., they would be hurting themselves via the value chain).
- **FVA is negative:** if domestic production uses more foreign inputs, TTBS protection is lower (protection rents leak abroad / domestic objective doesn't strongly value foreigners).
- **IP is positive:** consistent with domestic political economy: higher domestic production relative to imports correlates with higher TTBS use (more protection).

8 Quantitative model (for counterfactuals)

8.1 Preferences and demand

Consumers have quasi-linear preferences over a homogeneous good and an Armington composite:

$$U(d_0^h, \bar{d}^h) = d_0^h + \delta^h (d^h)^\psi, \quad \psi < 1,$$

with CES composite:

$$d^h = \left(\sum_{c=1}^C (d_c^h)^{(\sigma-1)/\sigma} \right)^{\sigma/(\sigma-1)}. \quad (3.1)$$

σ is elasticity across sources; ψ controls elasticity for the composite.

8.2 Roy–Fréchet mechanism for supplying GVC inputs

Each agent in country h owns a quantity of a GVC input and draws productivity for supplying it to each destination c :

$$F_h^c(z) = \exp(-A_h^c z^{-\theta}), \quad \theta > 1.$$

Probability an agent supplies to destination c :

$$\pi_h^c = \Pr \left(\bar{r}_h^c z_h^c \geq \max_{d \neq c} \{ \bar{r}_h^d z_h^d \} \right). \quad (3.2)$$

Total effective units of input supplied from h to c :

$$\bar{\nu}_h^c \equiv N^h \pi_h^c \mathbb{E} \left[z_h^c \mid \bar{r}_h^c z_h^c \geq \max_{d \neq c} \{ \bar{r}_h^d z_h^d \} \right]. \quad (3.3)$$

Income. Aggregate income includes labor, GVC income, tariff revenue, and transfers:

$$I^h = w^h L^h + \sum_c \bar{r}_h^c \bar{\nu}_h^c + R^h + B^h, \quad R^h = \sum_{c \neq h} \left(\frac{t_c^h}{1 + t_c^h} \right) p_c^h d_c^h.$$

8.3 Production

Competitive producers in country h produce a differentiated good using labor and a CES aggregate of GVC inputs from different origins:

$$q^h = z^h \left(\sum_{c=1}^C (\nu_c^h)^{(\varrho-1)/\varrho} \right)^{\alpha \varrho / (\varrho-1)} (l^h)^{1-\alpha}. \quad (3.4)$$

ϱ is the elasticity across input origins; α is the share of GVC inputs.

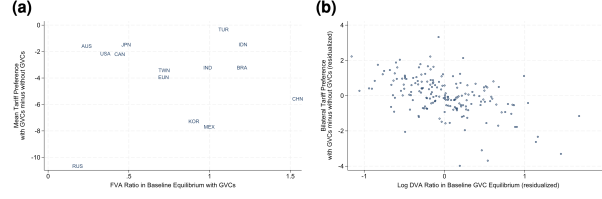


FIGURE 3
Increasing GVC input trade costs to impose GVC autarky. (a) Changes in mean tariff preferences and (b) changes in bilateral tariff preferences

Iceberg frictions for GVC inputs. If $\kappa_c^h \geq 1$:

$$r_c^h = \kappa_c^h \bar{r}_c^h, \quad \bar{\nu}_c^h = \kappa_c^h \nu_c^h.$$

8.4 Government problem in the quantitative model

Government objective keeps the same political-economy structure:

$$G^h = V^h + \delta^{DPE} \pi^h + \delta^{DVA} DVA^h + \delta^{FVA} FVA^h,$$

choosing tariffs subject to MFN constraints and equilibrium conditions.

8.5 Useful implied supply elasticity

They show bilateral factor supply can be written as:

$$\bar{\nu}_h^c = \Gamma \left(1 - \frac{1}{\theta} \right) A_h^c \left(\frac{\bar{r}_h^c}{\Phi^h} \right)^{\theta-1} N^h, \quad (3.5)$$

so $\partial \ln \bar{\nu}_h^c / \partial \ln \bar{r}_h^c = \theta - 1$. *Interpretation:* θ governs how “specific” inputs are to destinations; as $\theta \rightarrow 1$, this approaches strong specificity.

8.6 Calibration choices I note

They calibrate to 2005 data in a composite sector. They set:

$$\sigma = \varrho = 4, \quad \alpha = 0.9, \quad \psi = 0.5.$$

They assume $\delta^{FVA} = 0$ (domestic government doesn’t weight foreigners’ rents much), and match moments to pick:

$$\delta^{DPE} = 1, \quad \delta^{DVA} = 9.$$

(So the model needs substantial political weight on domestic upstream suppliers to match preference patterns.)

9 Figures 3–4: Quantitative counterfactuals

9.1 Figure 3: imposing “GVC autarky” by raising GVC input trade costs

Experiment. Increase GVC input trade costs to eliminate input trade (conceptually $\hat{\kappa} \rightarrow \infty$). MFN constraints are held constant, so changes in optimal tariff preferences correspond to changes in applied tariffs.

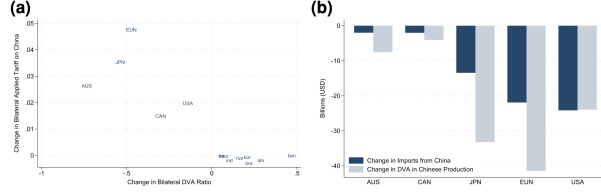


FIGURE 4
Decoupling China from G7 supply chains. a) Change in bilateral applied tariffs on China versus change in bilateral DVA ratio and b) changes in imports from China and domestic value added in Chinese Production

What the paper reports.

- Global mean applied tariff rises from about 5% (baseline) to 9.25% (GVC autarky).
- Mean tariff preference falls from about 4.9 percentage points (baseline) to 0.7 percentage points (autarky): preferences largely vanish.

Figure 3(a) interpretation. They plot the (baseline minus autarky) change in mean preferences against each importer’s FVA ratio in the baseline. The negative slope means: countries that rely more on foreign content in domestic production get the biggest preference reductions when GVCs are shut down. That fits the FVA channel: FVA is a key reason they kept tariffs low in the first place.

Figure 3(b) interpretation. They plot (baseline minus autarky) changes in bilateral tariff preferences against the baseline DVA ratio (both residualized for multilateral forces and imports). Negative association means: bilateral pairs with high domestic content embedded in partner exports are exactly the pairs where tariffs rise the most when GVCs are removed.

9.2 Figure 4: “decoupling” China from G7 upstream input supply

Experiment. Restrict G7 countries from supplying GVC inputs to China (set China-facing upstream productivity parameters to zero for those sources).

Figure 4(a) interpretation. The x-axis is the change in the bilateral DVA ratio vis-à-vis China; the y-axis is the change in the applied tariff on China. G7 countries experience large drops in DVA ratios and raise tariffs on China. **Policy complementarity:** upstream restrictions that cut GVC linkages to China make downstream tariffs on Chinese final goods more attractive.

Figure 4(b) interpretation. They decompose why DVA ratios change: both DVA in Chinese production and imports from China fall (China becomes higher cost without G7 inputs), but the relative magnitudes differ across G7 countries; that heterogeneity drives different DVA-ratio changes and tariff responses.

10 Conclusion

GVC linkages create a wedge between *the origin of a good* (where tariffs apply) and *the nationality of value-added income*. That wedge weakens classic mercantilist incentives for protection: tariffs hurt domestic upstream income when imports contain domestic content (DVA channel), and tariffs’ benefits leak to foreigners when domestic goods embody foreign inputs (FVA channel). The data support both forces: countries set lower preferential tariffs and use fewer temporary trade

barriers where GVC linkages are strongest. In counterfactuals, eliminating GVCs wipes out tariff preferences, and targeted decoupling from China raises optimal G7 tariffs on Chinese exports—so restricting upstream links and raising downstream tariffs can be complements in the policy calculus.

Takeaway. Global value chains (GVCs) make tariffs “self-harming” in new ways: when production is internationally shared in value-added terms, the classic terms-of-trade logic for protection is weakened, so optimal (discretionary) protection on *final goods* falls where GVC linkages are strongest.

What the paper asks. We can summarize the core questions as: (i) How do GVC linkages change a government’s incentive to impose import protection? (ii) Do these forces show up in real-world trade policy (tariffs and temporary trade barriers)?

Core. The key conceptual move is to stop thinking in “gross trade flows” only, and instead track *who earns the income embedded in goods* (value added). Because tariffs are levied based on *where goods are produced* (location), but income is earned based on *whose value added is inside the good* (nationality), GVCs drive a wedge between the object being taxed and the people ultimately affected.

Two mechanisms that push tariffs down. The paper emphasizes two robust channels for *final goods* tariffs:

1. **Domestic content in foreign final goods (“my VA is inside your exports”) reduces my incentive to tax you.** If Home supplies upstream value added used in Foreign’s production of the final good, then a Home tariff that lowers the foreign producer price also lowers the return to Home’s own upstream suppliers. So the usual “shift costs onto foreigners via the terms of trade” motive is muted.
2. **Foreign content in domestic final goods (“your VA is inside my import-competing production”) reduces my incentive to protect.** If Home’s own import-competing final-good production uses foreign value added, then a Home tariff that raises the local final-good price generates rents that leak upstream to foreign input suppliers. Protection becomes less attractive because some of the benefits are exported.

Theoretical prediction in words. Holding other forces fixed, discretionary tariffs on final goods should be *decreasing* in: (i) the domestic value-added content embodied in foreign-produced final goods, and (ii) the foreign value-added content embodied in domestically produced final goods.

How the paper connects theory to data. Empirically, they operationalize the theory using:

- bilateral *applied* tariffs vs. MFN tariffs (tariff preferences are applied minus MFN, censored above at 0 due to MFN rules);
- temporary trade barriers (TTBs: antidumping, safeguards, countervailing duties), measured via import coverage ratios;
- value-added content measures from input–output data (WIOD), mapped to industries and final-goods categories.

Empirical conclusions (what the data say). The read of the empirical results is:

- **Tariff preferences:** Countries offer *larger* tariff preferences (lower applied tariffs relative to MFN) exactly where theory says they should: when imports contain more of the importer’s own value added (higher “domestic content in foreign goods”), and when domestic production contains more foreign value added.
- **Institutional constraint matters:** Accounting for MFN censoring strengthens the estimated relationship (i.e., applied tariffs are less responsive than “optimal” latent tariffs because MFN acts as a ceiling on discriminatory tariffs).
- **Heterogeneity consistent with pass-through logic:** The negative relationship between tariffs and domestic value added is stronger when upstream inputs are more differentiated (interpreted as stronger pass-through from downstream prices to upstream returns).
- **TTBs:** Use of temporary trade barriers is also lower where GVC linkages are stronger. The sensitivity to domestic content is especially pronounced for cases with China as exporter, consistent with the idea that importers avoid actions that would boomerang onto their own upstream suppliers embedded in Chinese exports.

Quantitative conclusions (counterfactuals). The quantitative model is used to ask: *how big* is the effect of GVCs on tariffs?

- **GVC autarky:** If GVC input trade is effectively shut down, preferential tariffs largely disappear and average applied tariffs rise sharply. (In the paper’s calibration, mean applied tariffs rise from about 5% to about 9.25%, and mean tariff preferences collapse from about 4.9pp to about 0.7pp.)
- **Decoupling China from G7 supply chains:** Cutting off G7 upstream input supply to China reduces G7 value added embodied in Chinese exports, which then makes it optimal for G7 countries to raise tariffs on Chinese final goods. This implies **policy complementarity**: upstream restrictions on supplying China can raise downstream incentives to tariff China.
- **Historical implication (1995–2015):** Falling GVC frictions over time imply a sizable decline in optimal tariffs; the paper reports that changing GVC frictions can account for about a 2 percentage point reduction in global mean applied tariffs, which is meaningful relative to the observed decline in average tariffs over the period.

So what is the conclusion? The paper concludes that GVC linkages are a first-order determinant of modern trade protection on *final goods*: when production is internationally fragmented, governments internalize more cross-border incidence of tariffs (via value-added income), so they choose *less* protection and are more restrained in discretionary instruments like TTBs. Conversely, policies that sever GVC ties (especially targeted “decoupling”) can mechanically strengthen incentives for higher tariffs.

The paper’s contribution.

- A tractable “value-added” characterization of optimal tariff incentives with GVCs (income incidence is the organizing object).
- Clear, signable predictions for *final goods* tariffs that can be taken to data.
- Evidence across tariffs *and* TTBs that aligns with those predictions.

- Quantitative counterfactuals showing that removing GVC linkages changes not just trade volumes but also the *structure of protection* (especially tariff preferences).

A key boundary is that predictions for *input tariffs* are much less general and depend on additional structure (e.g. bargaining, hold-up, complementarity/substitutability of inputs). So the paper focuses empirically on final goods protection and leaves input-tariff determination as fertile future work.